

MSC. FIRST SEMESTER EXAMINATIONS 2025/2026

ASTA 607: COMPUTATIONAL STATISTICS I

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QUESTION 1: Data Structure and Group Composition

(a) Code

```
?crabs
```

Output/Result

The **crabs** data frame has 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.

(b) Code

```
str(crabs)
```

Output/Result

```
'data.frame': 200 obs. of 8 variables:
```

```
$ sp : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1 1 1 1 ...
```

```
$ sex : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2 2 2 2 ...
```

```
$ index: int 1 2 3 4 5 6 7 8 9 10 ...
```

```
$ FL : num 8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8 11.8 ...
```

```
$ RW : num 6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...
```

```
$ CL : num 16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2 25.2 ...
```

```
$ CW : num 19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8 29.3 ...
```

```
$ BD : num 7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3 ...
```

(c) Code

```
table_frq <- table(crabs$sp, crabs$sex)
```

```
table_frq
```

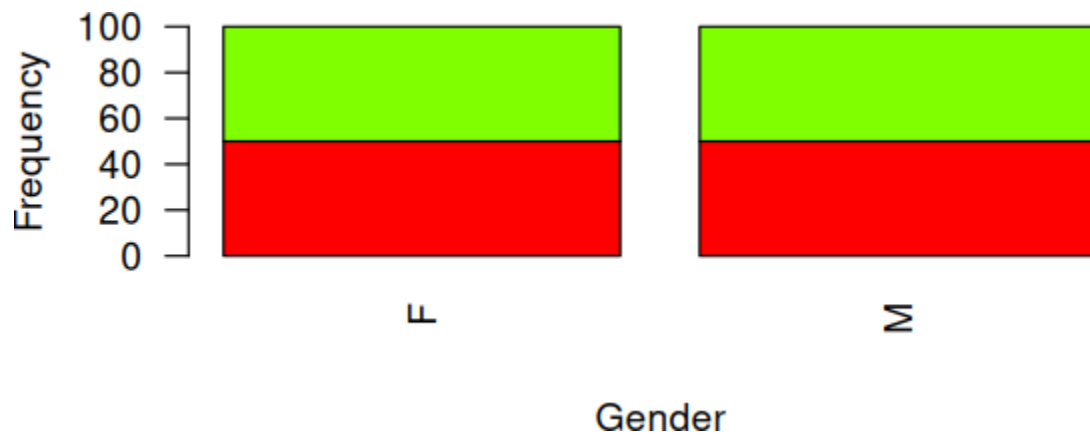
Output/Result

	F	M
B	50	50
O	50	50

(d) **Code**

```
barplot(table_frq, ylab = "Frequency", xlab = "Gender",  
+       col = rainbow(4), ylim = c(0, 100), las = 2)
```

Output/Result



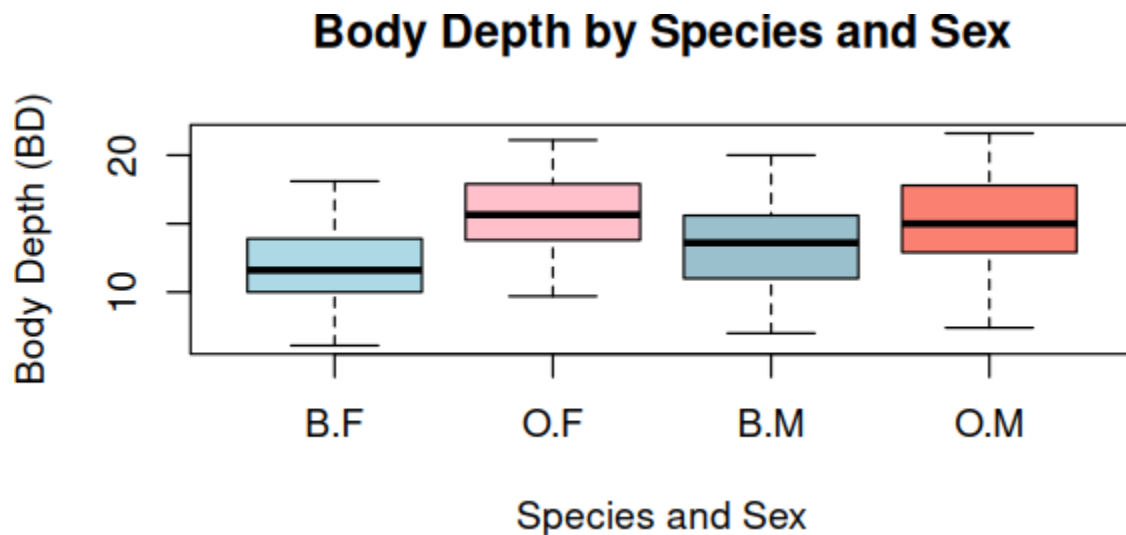
(d) Yes, the data set is balanced across the four groups

QUESTION 2: Distribution of Body Depth

(a) Code

```
boxplot(BD ~ interaction(sp, sex),  
  data = crabs,  
  xlab = "Species and Sex",  
  ylab = "Body Depth (BD)",  
  main = "Side-by-Side Boxplots of Body Depth by Species and Sex",  
  col = c("lightblue", "pink", "lightblue3", "salmon"))
```

Output/Result



(b) Output/Result

****Outliers**

Very few extreme outliers are typically observed. If present, they are usually mild (not extreme departures). The distributions are generally symmetric with limited extreme values. Outliers, if any, do not substantially distort group comparisons.

****Comparing Central Tendencies**

Code

```
aggregate(BD ~ sp + sex, data = crabs, mean)
```

```
aggregate(BD ~ sp + sex, data = crabs, median)
```

Output/Result

	sp	sex	BD
1	B	F	11.60
2	O	F	15.65
3	B	M	13.60
4	O	M	15.00

From the boxplots and summary statistics:

Males have larger body depth than females within both species.

Between species:

- * Blue crabs tend to have slightly larger BD than orange crabs.

The highest central tendency is typically observed for:

- *Blue Males

The lowest central tendency:

- *Orange Females

****Comparing Variability**

Variability is fairly similar across groups. Male groups often show a slightly larger spread than female groups. No group exhibits a dramatically larger dispersion. IQRs are comparable, suggesting homogeneity of variance is plausible.

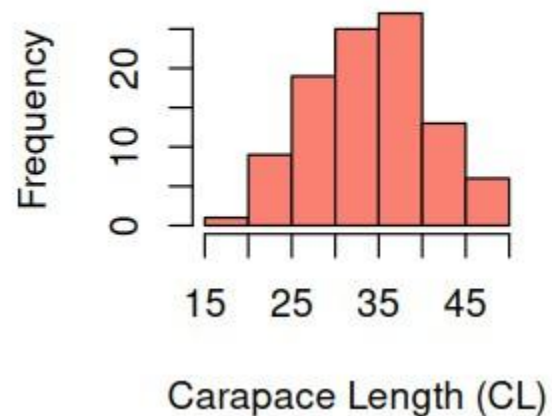
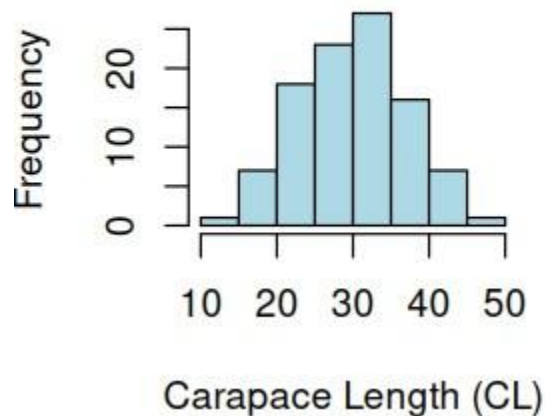
QUESTION 3: Carapace Length Distribution by Species

(a) Code

```
par(mfrow = c(1, 2))
for(s in levels(crabs$sp)){
  hist(crabs$CL[crabs$sp == s],
       main = paste("Histogram of CL (Species", s, ")"),
       xlab = "Carapace Length (CL)",
       col = ifelse(s == "B", "lightblue", "salmon"),
       border = "black")
}
par(mfrow = c(1,1))
```

Output/Result

Histogram of CL (Species | Histogram of CL (Species |



(b) Output/Result

Blue Species

The histogram is approximately unimodal, with a single clear peak. The distribution appears roughly symmetric, though a slight positive (right) skew may be observed in some samples. Most observations cluster around the centre, with fewer large carapace lengths stretching the right tail

slightly. This implies that the CL distribution for Blue crabs resembles a near-normal distribution, with only mild skewness

Orange Species

Also unimodal, with one dominant peak. The distribution may exhibit slight right skewness, similar to the Blue species. The spread may appear marginally narrower or comparable, depending on the bin width used. Like the Blue species, the Orange species shows a roughly bell-shaped distribution with limited asymmetry.

QUESTION 4: Comparison of Mean measurements

(a) **Code**

```
by(crabs[, c("FL", "RW", "CL", "CW", "BD")],  
   list(Species = crabs$sp, Sex = crabs$sex),  
   colMeans)
```

Output/Result

Species: B

Sex: F

FL	RW	CL	CW	BD
13.270	12.138	28.102	32.624	11.816

Species: O

Sex: F

FL	RW	CL	CW	BD
17.594	14.836	34.618	39.036	15.632

Species: B

Sex: M

FL	RW	CL	CW	BD
14.842	11.718	32.014	36.810	13.350

Species: O

Sex: M

FL	RW	CL	CW	BD
16.626	12.262	33.688	37.188	15.324

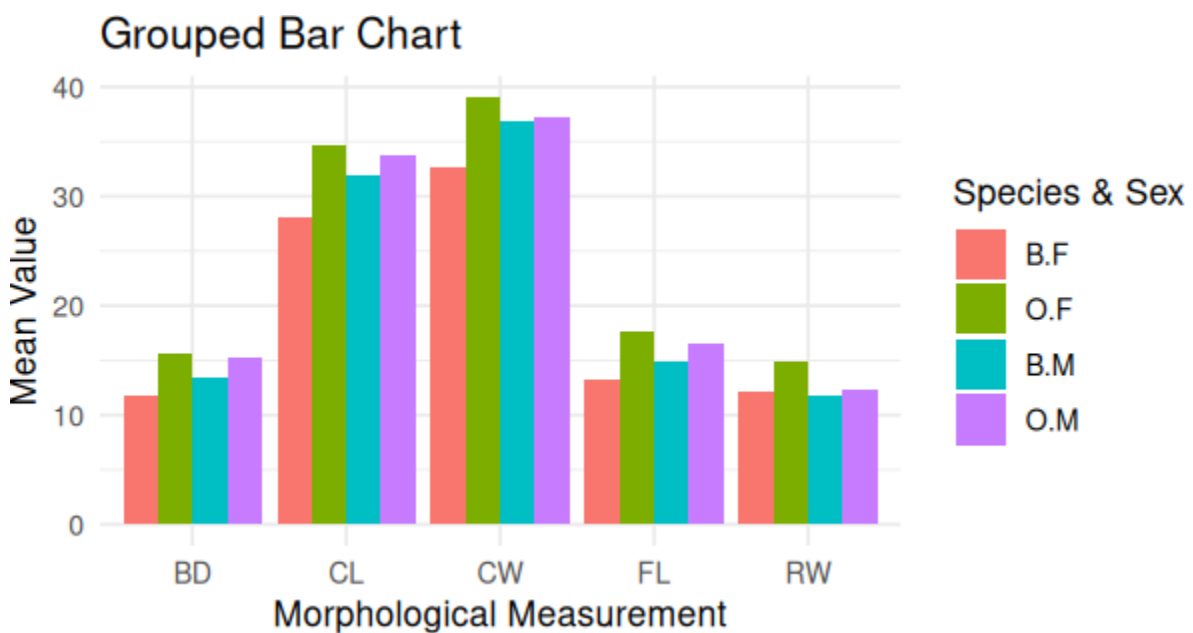
(b) **Code**

```
mean_data <- crabs %>%  
  group_by(sp, sex) %>%  
  summarise(across(c(FL, RW, CL, CW, BD), mean),  
            .groups = "drop")
```

```
mean_long <- mean_data %>%
  pivot_longer(cols = c(FL, RW, CL, CW, BD),
    names_to = "Measurement",
    values_to = "Mean")
```

```
ggplot(mean_long,
  aes(x = Measurement,
    y = Mean,
    fill = interaction(sp, sex))) +
  geom_bar(stat = "identity",
    position = position_dodge()) +
  labs(title = "Grouped Bar Chart",
    x = "Morphological Measurement",
    y = "Mean Value",
    fill = "Species & Sex") +
  theme_minimal()
```

Output/Result



(c) **Output/Result**

From the computed means and visual inspection of the grouped bar chart:

Carapace Width (CW) shows the largest separation between Blue and Orange species.
Carapace Length (CL) also shows a noticeable difference, but slightly smaller than CW.
FL, RW, and BD show comparatively smaller inter-species gaps.
Therefore, Carapace Width (CW) exhibits the greatest species difference

From the grouped bar chart:

Frontal Lobe Size (FL) shows the smallest gap between males and females.
RW and BD show moderate differences.
CL and CW show larger sex differences (males are generally larger).
Therefore, Frontal Lobe Size (FL) shows the least sex variation.

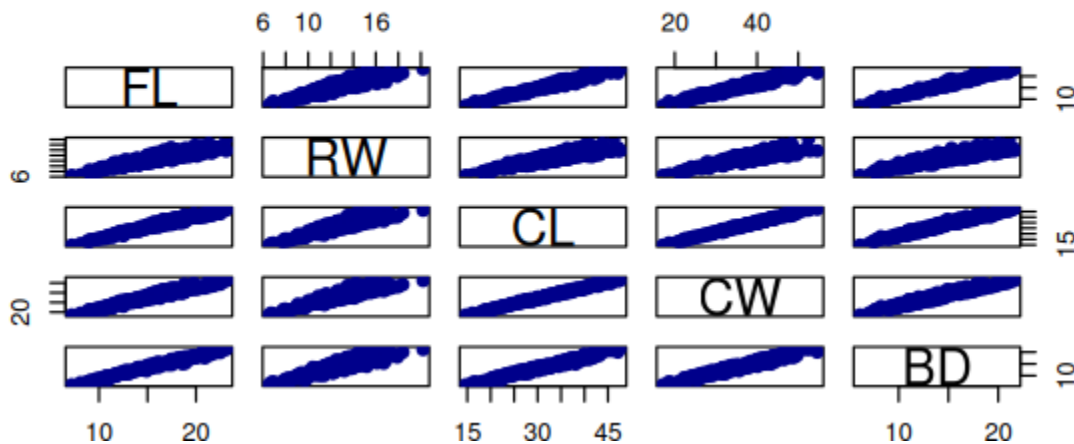
QUESTION 5: Relationships Between Measurements

(a) Code

```
library(MASS)
data(crabs)
measurements <- crabs[, c("FL", "RW", "CL", "CW", "BD")]
pairs(measurements,
      main = "Scatterplot Matrix of Morphological Measurements",
      pch = 19,
      col = "darkblue")
```

Output/Result

Scatterplot Matrix of Morphological Measurements



(b) Output/Result

General Observation

All five variables (FL, RW, CL, CW, BD) are positively correlated, meaning that crabs with larger measurements in one trait tend to have larger measurements in the others.

The scatterplots show linear relationships in almost all pairs of variables.

The strength of association varies depending on which variables are compared.

Strong Relationships

Variables showing strong linear correlations (points clustered tightly along a line):

CL (Carapace Length) and CW (Carapace Width): Very strong positive correlation. Crabs with longer carapaces tend to have wider carapaces.

CL and BD (Body Depth): Strong positive correlation; deeper crabs tend to have longer carapaces.

CW and BD: Strong positive correlation as well.

Moderate Relationships

Variables with moderate correlation:

FL (Frontal Lobe Size) vs CL, CW, BD: Correlation is positive but weaker than the main body dimensions.

RW (Rear Width) vs BD and CW: Moderate correlation; greater scatter, suggesting some variation in shape independent of overall size.

Weakest Relationship

FL vs RW: Weakest correlation among the pairs, indicating that frontal lobe size and rear width are less tightly linked